Avi Punjabi

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EDUCATION

University of California, Berkeley, *Berkeley, California*, 08/2026 – 05/2027

Master of Engineering in Mechanical Engineering, Concentration in Control of Robotic and Autonomous Systems

Texas A&M University, *College Station, Texas*, 08/2022 – 05/2026

Bachelor of Science in Mechanical Engineering, Minor in Computer Science

SKILLS

- Control System Design
- Robot Kinematics & Dynamics
- Embedded Systems Programming
- Data Acquisition & Visualization
- Kalman Filtering & Sensor Fusion
- CAN Bus, I2C, SPI, UART/Serial
- Reinforcement Learning
- MuJoCo, Docker, OpenCV
- C++, Python, ROS2, MATLAB
- Git & Linux, ESP32 & STM32
- Circuit Design & Signal Conditioning
- Testing & Validation Design
- SolidWorks, Siemens NX
- ANSYS, FEA, CFD, Thermal Analysis
- Mechanical Design, GD&T, FMEA

WORK EXPERIENCE

MRI & Robotics Mechanical Engineering Intern, Neuro42, *San Francisco, California,* 10/2026 – Present

- Designed and prototyped MRI scanner and neurosurgical robot components using CAD, drawings, and 3D printing.
- Assembled, integrated, and troubleshot mechanical, pneumatic, and electromechanical systems for validation.
- Collaborated with mechanical, electrical, embedded, and software teams to integrate MRI-compatible hardware.
- Ran hardware tests and documented failure modes, results, and troubleshooting steps to drive design iterations.
- Refined mechanical designs to resolve system integration challenges and improve overall product reliability.

Robot Dynamics Subteam Lead, SpaceCraft VR Lab, *College Station, Texas*, 08/2025 – 05/2026

- Derived forward and inverse kinematics for Canadarm2, the ISS robotic arm, mapping joint states to end-effector pose.
- Implemented position- and velocity-level control algorithms, enabling accurate closed-loop end-effector tracking.
- Integrated control algorithms into the Space Teams PRO simulator to validate performance in mission scenarios.
- Presented research poster to faculty and industry judges at the AggiE_Challenge Research Expo.

Mechanical Design Intern, Acutronic Actuation Inc., *Austin, Texas*, 05/2025 – 08/2025

- Modeled 20+ motor, actuator, gearbox, and test stand components, standardizing interfaces to cut rework.
- Ran FEA, thermal, and CFD simulations on 50+ assemblies in ANSYS to validate designs prior to fabrication.
- Partnered with machinists and engineers to evaluate manufacturability and ensure designs met tolerances.
- Automated test data analysis in Python, cutting analysis turnaround time for the design team.
- Documented testing and analysis procedures, improving reproducibility across the design team.

PROJECT EXPERIENCE

Mechanical Design Capstone Member, Space Kinetic Capstone Project, *Berkeley, California,* 09/2026 – Present

- Designing an automated payload reload mechanism with repeatable, precise alignment for a reusable spacecraft.
- Led trade studies on complexity, alignment accuracy, and robustness, selecting a design reusing existing actuation.
- Modeled the mechanism in SolidWorks and ran MATLAB motion, load, and tolerance analyses to verify alignment specs.
- Performed a Design FMEA of the reload sequence to identify jam, misalignment, and single-point failure risks.
- Presented resulting design changes to Space Kinetic's CTO and engineering experts in formal design reviews.

Data Acquisition & Processing Lead, Suspension Tester Capstone Project, *College Station, Texas*, 08/2025 – 05/2026

- Engineered a real-time DAQ system integrating IMUs, load cells, and strain gauges on a double-wishbone suspension.
- Wrote Teensy 4.1 firmware reading HX711 strain gauge amps, VL53L1X ToF sensors, and MPU-6050/LSM303DLHC IMUs.
- Built a Python live-monitoring dashboard fusing sensor streams for state estimation of suspension performance.
- Built and validated a single-wheel suspension tester applying controlled static and dynamic loads.
- Established test rig SOPs and safety protocols, and managed project budget and vendor pricing within funding.

Suspension Subteam Member, Texas A&M Solar Car Racing, *College Station, Texas*, 05/2023 – 05/2026

- Engineered 10+ braking and wheel assembly components, taking each from CAD modeling through simulation.
- Verified component performance against design requirements through physical testing.
- Presented and defended steering and brake hub designs in formal reviews, securing approval for fabrication.
- Analyzed braking system tradeoffs in manufacturability and cross-system impacts to guide component selection.
- Synthesized 50+ automotive engineering resources to inform brake system design decisions.

VEXU Team Lead; Treasurer; Underclassman Representative, Aggie Robotics, *College Station, Texas*, 08/2022 – 06/2025

- Led team through 2 international VEXU competitions, coordinating testing, logistics, and budgeting.
- Modeled and simulated 40+ robot components in CAD/FEA to verify structural integrity before fabrication.
- Directed development of a two-axis motorized flywheel turret assembly for competition robots.
- Applied a hybrid Gantt/Agile framework to keep turret design and build milestones on schedule.
- Restructured testing to run concurrently with hardware and software development, shortening iteration cycles.